

Worldwide Population Study on Patterns of Human Immunodeficiency Virus (HIV) and Tuberculosis (TB) Cases

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May 26, 2017

Abstract

This paper took a multidisciplinary approach of combining the fields of bacteriology, virology and epidemiology. We assessed the epidemiological consequences of both Human Immunodeficiency Virus (HIV) and Tuberculosis (TB) using graphical representations and various statistical analyses through Python 3.5 programming via the utilization of datasets about the worldwide occurrence of TB and HIV. Granted this, it was then hypothesized that HIV increases the chance of an individual getting TB and extrapulmonary TB (EPTB), thus increasing affected individuals' chance of dying. Furthermore, socio-economic conditions of various nations were surveyed to determine possible risk factors.

Keywords

Human Immunodeficiency Virus (HIV), Tuberculosis (TB), extrapulmonary TB (EPTB), epidemiology, co-infection

Introduction

Tuberculosis is a communicable infectious disease, infecting approximately one third of the world's population¹. It is caused by *Mycobacterium tuberculosis*, an obligate pathogenic bacterium which can not only infect individuals with pulmonary tuberculosis (TB), but also extra-pulmonary tuberculosis (EPTB)². TB occurs when *M. tuberculosis* is solely affecting the lungs, while EPTB occurs when the bacteria infect other organs such as the lymph nodes, bone, central nervous system, and pericardium³.

Although there are many risk factors for TB, the chances of the disease developing in an individual is highly dependent on their immunocompetence. Notably, one of the preeminent causes of immunosuppression is the Human Immunodeficiency Virus (HIV), a retrovirus that specifically targets the CD4⁺ T cells which are vital to fight off pathogens like *M. tuberculosis*⁴.

TB is the most common disease as well as the leading cause of death among HIV-infected individuals⁵. In addition, 40% to 80% of individuals co-infected with TB and HIV also have EPTB, while only 10% to 20% of individuals without HIV acquire EPTB⁶. Hence, HIV-infected patients are more susceptible to not only pulmonary TB, but also EPTB.

Numerous studies have pointed towards a fundamental correlation between HIV and TB as well as HIV and EPTB. It is known that HIV substantially increases the chance of getting TB; however, there have been relatively few macro-scale epidemiological studies focusing particularly on epidemiology of HIV and TB co-occurrence. Consequently, this paper scrutinizes the prevalence pattern of TB,

EPTB and HIV across the globe using worldwide data from several different years in order to assess the hypothesis of how an individual with HIV has an increased chance of acquiring both TB and EPTB.

Materials & Methods

Dataset

Databases from World Health Organization (WHO) were obtained to collect international data on patients affected with HIV, TB, and both diseases. There was HIV and TB data for a total of 106 countries in 2015 and 105 countries in 2010, 2005, and 2000 each. There was also HIV and EPTB data for a total of 106 countries in 2015, 105 countries in 2010, 102 countries in 2005, and 98 countries in 2000.

Programming Language

Python 3.5 programming language was used to create figures and for analysis. Through Python 3.5, several statistical libraries were used to analyze the data in csv files. Pandas was used to read csv files and conduct data analysis, while Plotly was used for interactive visualization and to generate various graphs. An online Jupyter Notebook was used through IBM's Data Science Experience in order to share the work among group members.

Data Analysis

Various types of figures were employed including line graphs, scatter plots, bar graphs, and a world map showing prevalence of total cases of TB patients with HIV. Statistical test values were obtained through Scipy's built-in linear regression function which calculated Pearson's correlation coefficient (r) and p -value. Cochran-Armitage test for trend was done using Excel and XLSTAT.

Results

To demonstrate the relationship between TB and HIV, the TB cases per 100k people were compared to the HIV cases per 100k people for each individual country. These data were collected for the following four years: 2000, 2005, 2010, and 2015. Subsequently, a scatter plot was used to combine the aforementioned HIV and TB cases for all countries as shown in Figure 2. Statistical analyses on the scatter plots for the four separate years were conducted by extrapolating the line of best fit, where the R , R^2 , and p -values were then determined. All together, the R^2 values ranged from 0.44 to 0.66 ($p < 0.05$).

To compare the numbers of TB cases and HIV cases with individuals co-infected with TB and HIV ($TB^+ HIV^+$), bar graphs were generated for all the four years. First, a bar graph to compare the values of the previously mentioned variables for five of the top HIV countries versus five of the lowest HIV countries were developed for each of the four years as shown in Figure 3a. Second, in order to better visualize and compare the number of TB, HIV and $TB^+ HIV^+$ cases, separate bar graphs were formed for the top HIV countries and the low HIV countries as shown in Figures 3b and 3c.

To examine the relationship between HIV and EPTB, a scatter plot (Figure 4) was made, where the average R^2 value was determined to be 0.27 (average $p < 0.05$). Subsequently, as shown in Figure 5, a bar graph for some of the highest HIV and lowest HIV containing countries were made, where the

HIV and EPTB prevalence amounts were normalized in order to adjust the scale for the x-axis for an easier comparison.

In order to examine the wide scale effects of HIV on the total death toll of individuals who suffered from both HIV and TB for over the past 16 years (from 2000 to 2015), a line graph was delineated by combining the fatalities across an average of 174 countries as depicted in Figure 6. Notably, there was an overall increase in the total number of deaths by about 8.5% when individuals were co-infected with both diseases.

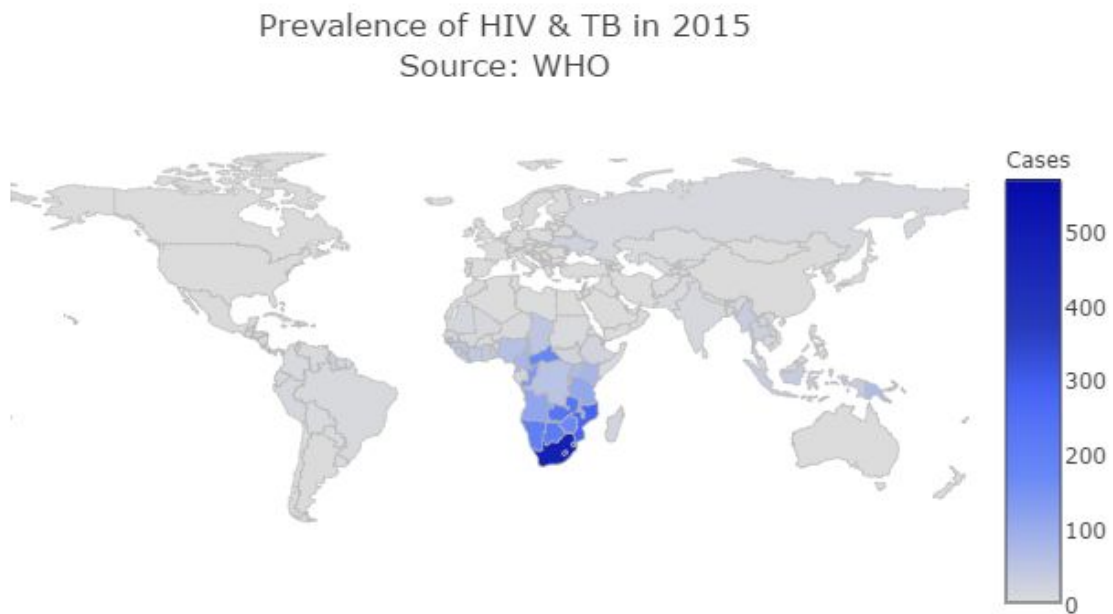
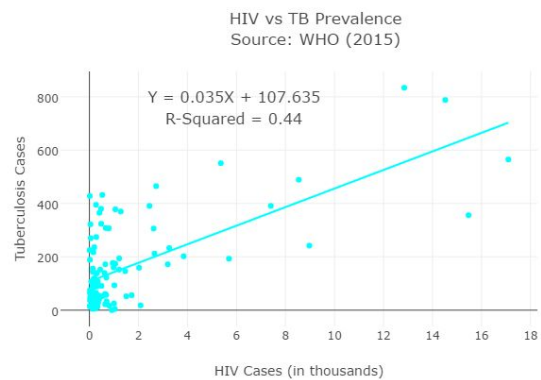
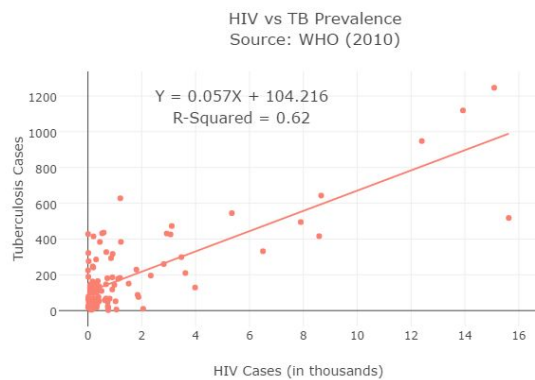
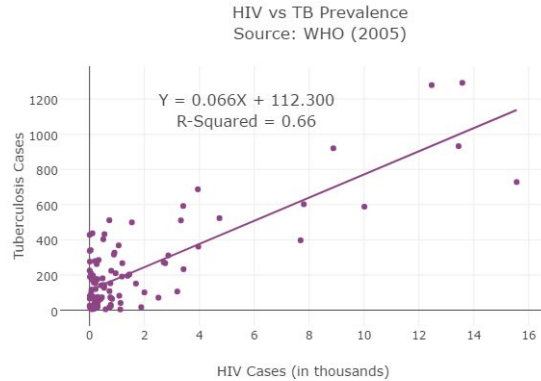
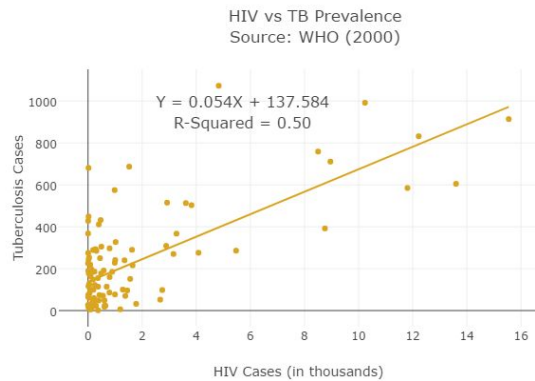


Figure 1. World map showing the distribution of TB⁺ patients infected with HIV (TB⁺ & HIV⁺). Evidently, there are greater number of HIV and TB co-infection cases in southern parts of Africa compared to other countries with collected data.

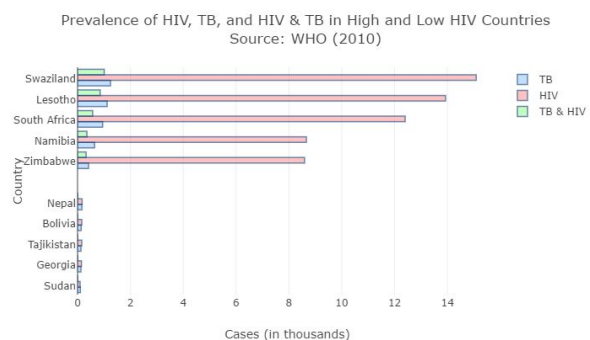
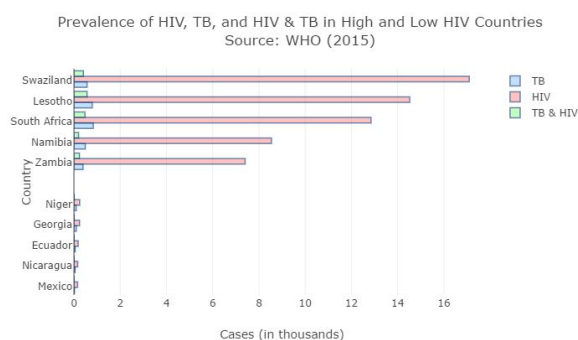
Year	Total Population	TB Cases	HIV Cases	TB & HIV Cases
2000	3 389 504 320	7 915 987	22 911 900	1 092 298
2005	3 679 784 138	8 468 439	25 009 700	1 282 534
2010	3 974 117 018	8 388 953	26 081 700	1 234 303
2015	4 281 947 174	8 207 281	28 280 900	1 011 965

Table 1. Total sample sizes (N) for total population, number of TB cases, HIV cases, and TB & HIV cases of all countries combined.



2015 (106 countries): $R^2 = 0.44$ | P-value = $8.3329877573e-15$
 2010 (105 countries): $R^2 = 0.62$ | P-value = $3.36874933079e-23$
 2005 (105 countries): $R^2 = 0.66$ | P-value = $1.14448184245e-25$
 2000 (105 countries): $R^2 = 0.50$ | P-value = $3.53707182784e-17$

Figure 2. The prevalence of HIV versus TB for 106 countries in 2015 and 105 countries in 2010, 2005, and 2000 each. All scatter plots reveal a positive correlation between HIV and TB cases i.e. as HIV prevalence (independent variable) increases, the TB prevalence (dependent variable) also increases.



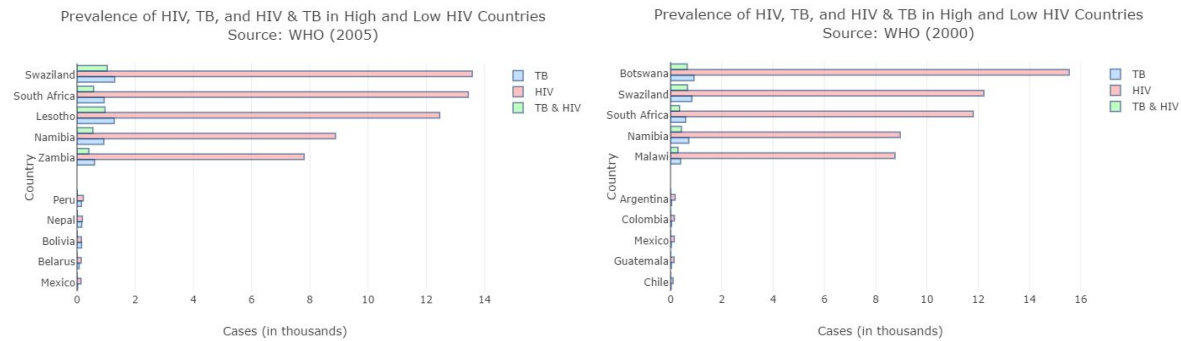


Figure 3a. Comparison of the prevalence of HIV, TB, and HIV & TB across 106 countries in 2015 and 105 countries in 2010, 2005, and 2000 that have high and low levels of HIV. Across all bar graphs, whereas the top 5 countries with the most HIV presence have a high amount of TB and TB & HIV presence, the bottom 5 countries with the least HIV presence have a low amount of TB and TB & HIV presence.

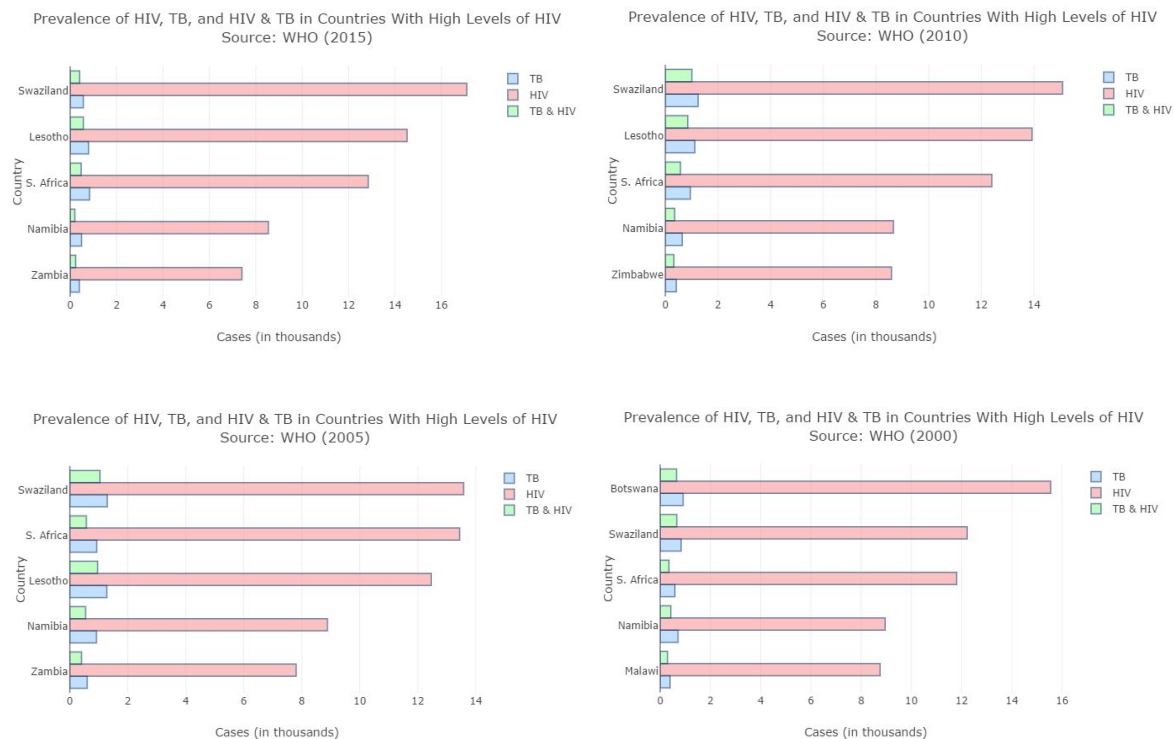


Figure 3b. Comparison of the prevalence of HIV, TB, and HIV & TB between the top 5 HIV prevalent countries. A closer look at the top 5 countries in Figure 4a to better show that high HIV presence correlates to high presence of TB and TB & HIV cases.

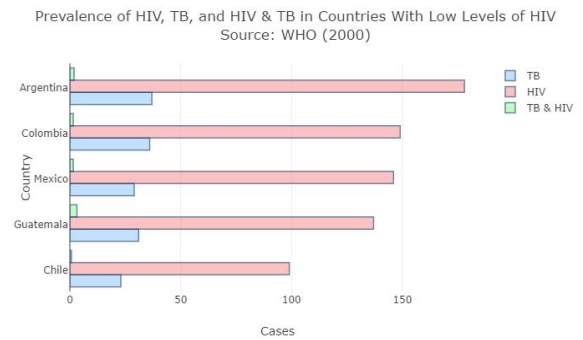
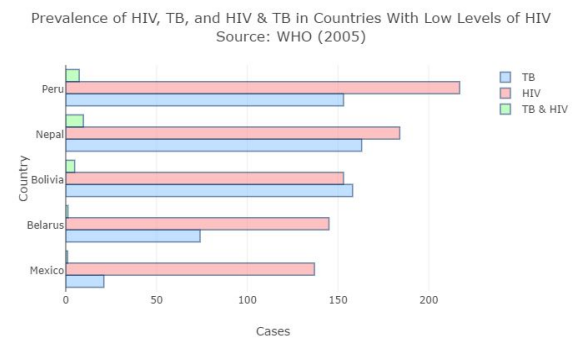
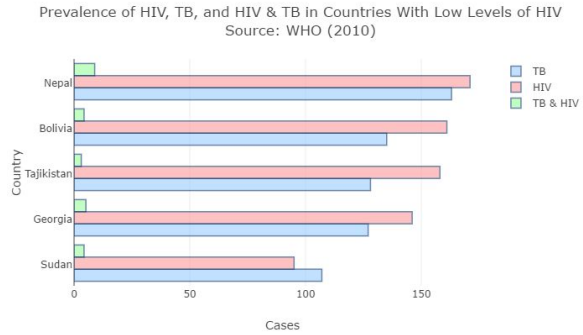
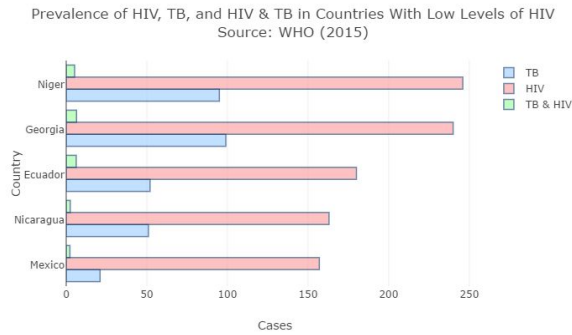
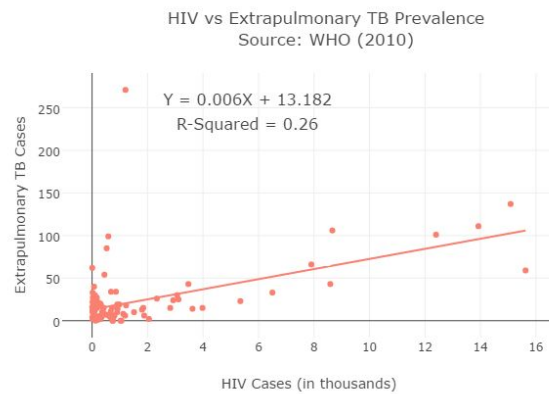
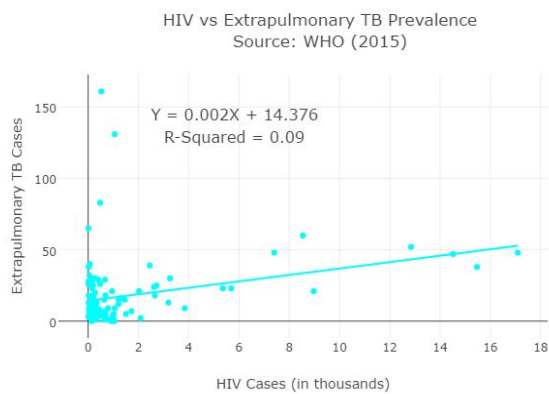
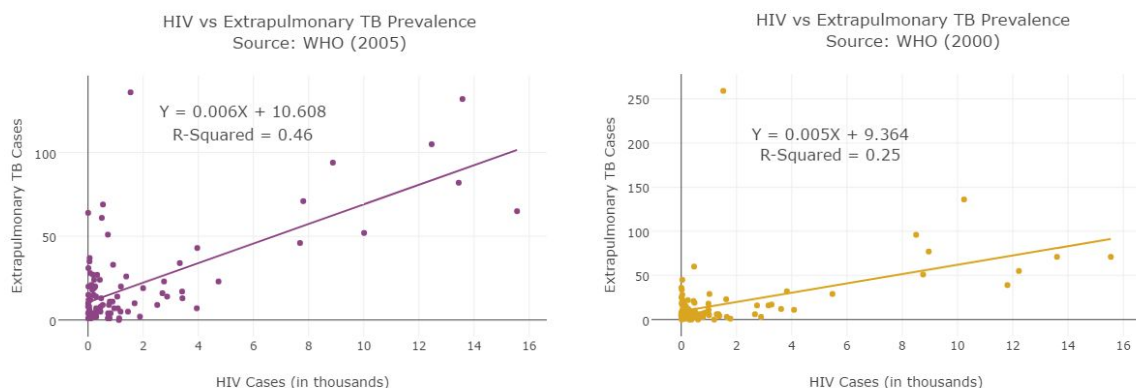


Figure 3c. Comparison of the prevalence of HIV, TB, and HIV & TB between the bottom 5 HIV prevalent countries. A closer look at the bottom 5 countries in Figure 4a to better show that low HIV presence correlates to low presence of TB and TB & HIV cases.





2015 (106 countries): $R^2 = 0.09$ | P-value = $1.85915942884e-03$
 2010 (105 countries): $R^2 = 0.26$ | P-value = $2.31924547247e-08$
 2005 (102 countries): $R^2 = 0.46$ | P-value = $5.16760484033e-15$
 2000 (98 countries): $R^2 = 0.25$ | P-value = $1.71807825004e-07$

Figure 4. The prevalence of HIV versus EPTB for 106 countries in 2015, 105 countries in 2010, 102 countries in 2005, and 98 countries in 2000. All scatter plots reveal a positive correlation between HIV and EPTB cases. i.e. as HIV prevalence (independent variable) increases, the EPTB prevalence (dependent variable) also increases.

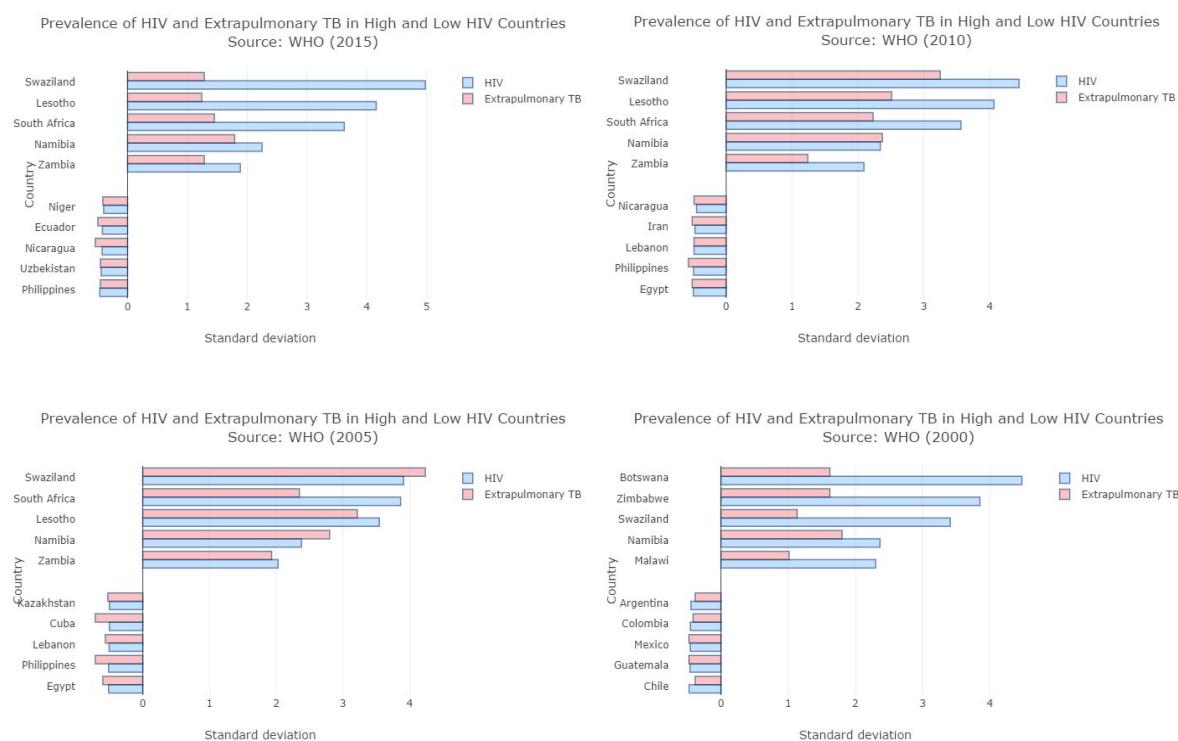


Figure 5. Comparison of the prevalence of HIV and EPTB across 106 countries in 2015, 105 countries in 2010, 102 countries in 2005, and 98 countries in 2000 that have high and low levels of HIV. Across all bar graphs, whereas the top 5 countries with the most HIV presence have an above average EPTB presence (positive standard deviation), the bottom 5 countries with the least HIV presence have a below average EPTB presence (negative standard deviation).

year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
country	157	157	158	163	168	174	173	172	174	180	185	183	184	183	185	181

Table 2. Number of countries used for each year for TB mortality rates graphs.

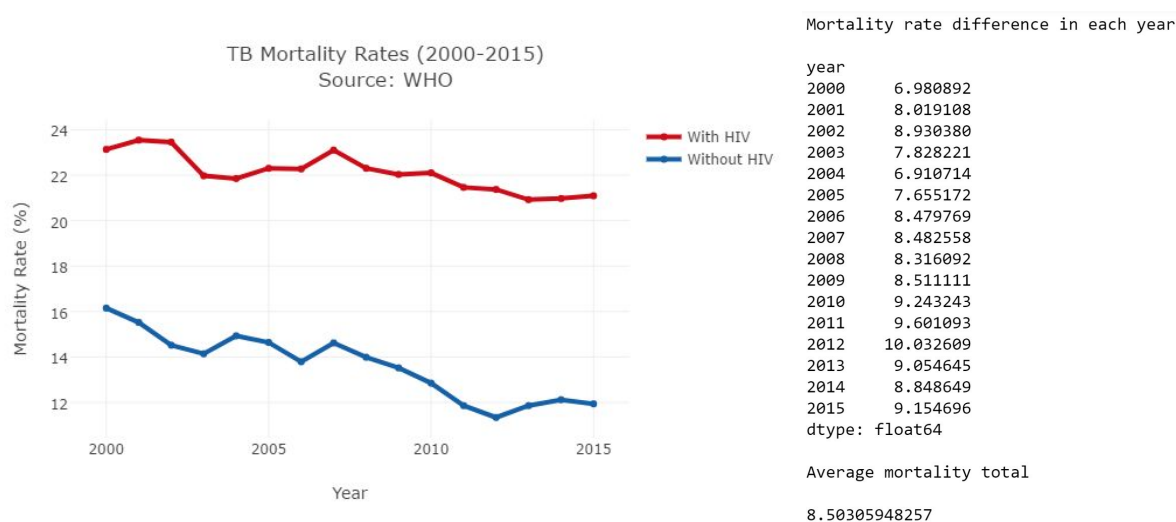
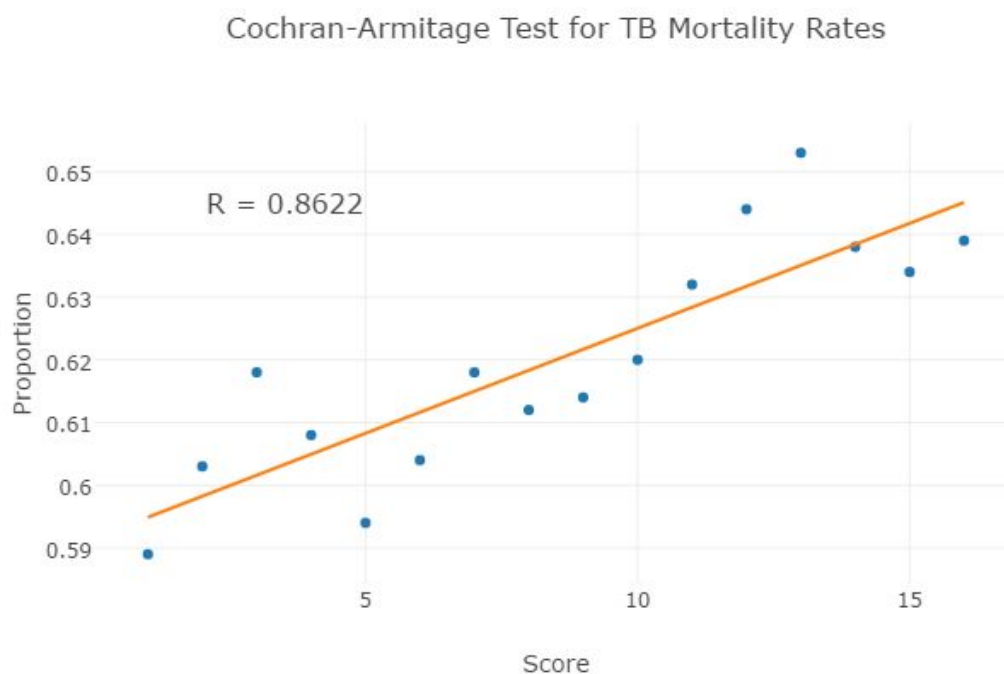


Figure 6. Comparison of the overall number of deaths due to TB in individuals with HIV versus those without HIV throughout all years from 2000 to 2015. Across all years, there is always an increase in total deaths for those who have HIV. The line graphs also show a downward trend of mortality rate over time for TB patients with and without HIV.



Cochran-Armitage trend test (Asymptotic p-value) / Two-tailed test:

|z| (Observed value) 7.568
|z| (Critical value) 1.960
p-value (Two-tailed) < 0.0001
alpha 0.05

Test interpretation:

H0: There is no association between the observed proportions and the score variable.

Ha: There is an association between the observed proportions and the score variable.

Cochran-Armitage trend test (Monte Carlo method - Number of simulations = 5000) / Two-tailed test:

|z| (Observed value) 7.568
|z| (Critical value) 0.025
p-value (Two-tailed) < 0.0001
alpha 0.05

Figure 7. The Cochran-Armitage trend test alongside the Monte Carlo method shows that a trend between TB mortality rates with and without HIV is present. The graph demonstrates that over the years, there exists an inverse correlation such that as the proportion of TB mortality rates with HIV increases, the proportion of TB mortality rates without HIV decreases. As the Cochran-Armitage trend test uses an approximation, the Monte Carlo method was used as an additional test for more exact results. Given that the p-values in both test results are much lower than their respective alphas, there is a strong evidence that such trend exists.

Discussion

Pearson's correlation coefficient (r), coefficient of determination (R^2), equation of the line of best fit, and p-value were used to test whether there was a strong and statistically significant correlation between the TB and HIV prevalences. The results indicated that in countries with higher HIV prevalence, there was also a higher TB and EPTB prevalence. On the other hand, the opposite was also shown as well, where in countries with lower HIV (i.e. lower HIV countries) prevalence, there were also lower prevalence of both types of TB. First, These findings were determined by the presence of a positive correlation in both scatter plots (Figures 2 & 4). This positive correlation can also be seen by using the slope of the line of best fit which was found to have positive values for all years. Notably, the value of the slope for the TB shows a strong correlation as it is close to having a slope value of 1 which would mean a one-to-one relationship between the independent and dependent variables.

Furthermore, the R^2 value was calculated and had an average value of 0.555 for TB and 0.265 for EPTB for the four years previously mentioned. The R^2 value of the TB line of best fit reveals that there is minimal variance between each individual data and the linear model. This reveals that there may be a relation between the two variables. On the other hand, a possible reason for why the EPTB R^2 value is lower than that of the TB can be due to the lack resources that there needs to be in order to detect the spread of TB to other organs^{7,8}. As mentioned in other studies, there are numerous reasons why the current diagnostic methods for EPTB are deficient which would account for the greater variability that exists between the data for the EPTB scatter plot. For instance, these diagnostic methods are invasive in nature and there is a lack of accuracy as in many cases, individuals who actually do have EPTB show negative results in their diagnosis due to the paucibacillary nature of EPTB^{7,8}.

Also, the p-values were determined to be less than 0.05 for both scatter plots. Subsequently, the null hypothesis would be that there is no correlation between HIV and TB prevalence. However, given that the p-values were less than 0.05, the null hypothesis was rejected. Thus, the hypothesis of this paper agrees with these statistical test findings as it was predicted that individuals who have HIV have a much higher chance of acquiring both TB and EPTB.

As mentioned previously, HIV is just one of the risk factors that would increase one's chance in getting TB. Other confounding variables or in this case, risk factors can potentially exist that would affect the high numbers of TB occurrences in the countries mentioned. Other risk factors that could help to account for the increase in TB in many of the mentioned countries include poverty and overcrowding. As shown in Figure 1, many of the countries with high TB prevalence exist in southern part of Africa where there is considerable poverty. Poverty is usually correlated with lack of treatment as well as education in various infections. According to a previous study, other risk factors include undernutrition and indoor air pollution⁹ which are also characteristic in many part of developing nations that were observed to have high prevalence of HIV⁺ & TB⁺ in Figure 1. Ultimately, most countries that had high HIV and TB prevalence were developing, while most developed nations showed a low number of cases for all HIV, TB, and HIV⁺ & TB⁺. This socio-economic implication signifies the need to target developing nations in reducing not only HIV and TB, but many other diseases as well.

Figure 6 showed a downward trend in mortality rate of TB patients with and without HIV overtime. This may imply that the health care and education throughout the world is becoming better every year. Furthermore, the line graph clearly showed that TB patients with HIV were much more likely to die than patients without HIV, ultimately showing the lethality of HIV using big data rather than biological mechanisms.

In order to confirm the existence of a trend between the mortality rates with and without HIV, the Cochran-Armitage trend test was used. From a graphical point of view, the test showed an inverse correlation whereby as the proportion of mortality rates with HIV increased over the years, the proportion of mortality rates without HIV decreased. In order to combat approximation, which is what the standard Cochran-Armitage computes, the Monte Carlo method was additionally used as an extra test to compute more accurate results. The null hypothesis would be that there is no trend present between mortality rates with and without HIV. However, as both trend tests exhibited p-values much less than the alpha (0.05), there is strong evidence of a trend existing with a risk of rejection being less than 0.01%.

Future direction: in order to support and prove better whether HIV increases EPTB specifically, future surveys could collect data for individuals who are both HIV and EPTB positive as the data from WHO were only collected from individuals who had either HIV or EPTB, unlike the data for TB. By collecting such data, one can more conclusively demonstrate whether HIV increases the risk of EPTB.

Acknowledgement

The authors would like to thank STEM Fellowship for providing many resources, workshops, and most importantly the Big Data Challenge for University of Toronto students. They would also like to

thank IBM for the Data Science Experience which was tremendously helpful as an online platform for collaboration.

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